

Birla Institute of Technology and Science, Pilani
Digital Signal Processing (ECE/EEE F434)

Comprehensive Examination, Closed Book, First Semester 2024 – 25

Maximum Time: 120 minutes

Date: 04/12/24

Maximum Marks: 50

Note: Answer all the questions sequentially and all parts of a question together.

1. Justify the following with reasoning, no marks for incorrect justification.
 - (a) Specify, if the sequence $x(n) = \{0, 1, 1, 0, 0, 0, -1, -1\}$ will have a real-valued or imaginary-valued 8-point DFT, explain without actually computing the DFT.
 - (b) A filter is obtained by cascading Type 2 linear phase FIR filter, $H_2[z]$, as $G[z] = H_2[z]H_2[z]$. The resulting filter $G[z]$ is of what Type?
 - (c) If the CT system is an all-pass system, its poles will be at locations s_k in the left-half of the s -plane and its zeros will be at corresponding locations $-s_k$ in the right-half of the s -plane. Which design method(s) out of impulse invariance and the bilinear transformation will result in an all-pass discrete-time system?
2. Similar to DFT, another popular transform is discrete cosine transform (DCT), which is used in image-coding applications. The DCT algorithm is developed via DFT as follows. Let $x(n)$ be a length N sequence defined for $0 \leq n \leq N-1$. First $x(n)$ is extended to length $2N$ by zero-padding and denoted as $x_e(n)$. Then a $2N$ length sequence is formed as $y(n) = x_e(n) + x_e(2N-1-n)$, $0 \leq n \leq 2N-1$ and its $2N$ -point DFT is computed and denoted as $Y[k]$. Then, the N -point DCT of $x(n)$ is defined as $C_X[k] = W_{2N}^{k/2} Y[k]$, $0 \leq k \leq N-1$. Show that

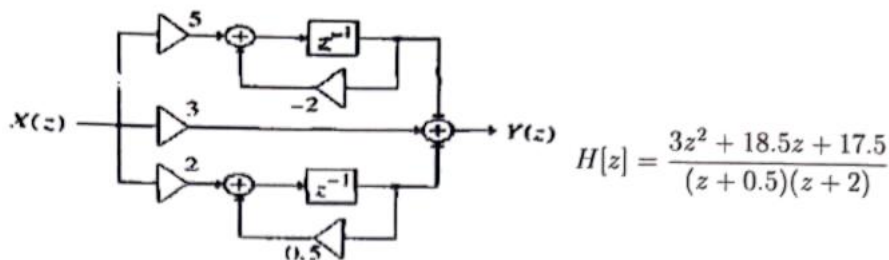
$$C_X[k] = \sum_{n=0}^{N-1} 2x[n] \cos\left(\frac{(2n+1)k\pi}{2N}\right), \quad 0 \leq k \leq N-1.$$

3. The Hann window is given as $w_{hn} = 0.5 \left[1 - \cos\left(\frac{2\pi n}{N-1}\right)\right]$, $0 \leq n \leq N-1$. You are given a new window $w(n)$ which is expressed as $w(n) = \left[\sin^4\left(\frac{\pi n}{N-1}\right)\right]$.
 - (a) How is this new window related to the Hann window in both time and frequency domains.
 - (b) Design a high pass linear phase and symmetric FIR filter with the desired frequency response given as

$$H_d(e^{jw}) = e^{-jw\tau}, \quad w_p \leq |w| \leq \pi.$$

For your design, take $N = 7$ and $w_p = 2$ rad/sec. Use Hann window.

4. Consider a linear-phase lowpass FIR filter whose coefficients are $h_1(k) = [-0.8, 1.6, 25.5, 47, 25.5, 1.6, -0.8]$, and whose DC gain, $H_1(0)$, is equal to 99.6. If we change those coefficients to $h_2(k) = [-0.8, 1.6, Q, 47, Q, 1.6, -0.8]$, we obtain a new DC gain equal to 103.6. What is the value of Q ?
5. You design an IIR digital filter with transfer function $H[z]$ and realize it as given below. During the final testing, you found that two of the multiplier coefficients in the realization have wrong values. Find these two multipliers and their correct values.



6. Design an IIR Chebyshev Type I digital filter to meet the following design specifications: passband ripple value between 1 and 0.8 with passband edge frequency 800 Hz, stop band ripple of 0.2 with stop band frequency of 2400 Hz. The sampling frequency being 8000 Hz. Find the transfer function of the analog filter. Use bilinear transformation to find the digital filter transfer function.

(???)

Hint: In general, the analog filter TF for one of the cases is given below, the other case can be deciphered from this, as discussed in the lecture.

$$H(s) = \frac{\Omega_p^N c_0 \prod_{k=1}^{\frac{N-1}{2}} c_k}{(s + \Omega_p c_0) \prod_{k=1}^{(N-1)/2} (s^2 + b_k \Omega_p s + c_k \Omega_p^2)},$$

where $c_0 = y_N$, $c_k = y_N^2 + \cos^2 \frac{(2k-1)\pi}{2N}$, $b_k = 2y_N \sin \frac{(2k-1)\pi}{2N}$, and $y_N = \frac{1}{2} \left[\left[\sqrt{1 + \frac{1}{\epsilon^2}} + \frac{1}{\epsilon} \right]^{1/N} - \left[\sqrt{1 + \frac{1}{\epsilon^2}} + \frac{1}{\epsilon} \right]^{-1/N} \right]$.

7. From a causal second order analog transfer function $H(s)$, you obtain a second order digital filter $G[z]$ by impulse invariance technique, show that $G[z]$ is given as

(3)

$$G[z] = \frac{z^2 - z \exp(-bT) \cos(aT)}{z^2 - 2z \exp(-bT) \cos(aT) + \exp(-2bT)} \text{ where } H(s) = \frac{(s+b)}{(s+b)^2 + a^2}.$$

For this part of the paper, marking scheme has not been defined, except for two parts. You are supposed to decipher and that gives you [6 marks (for solving first three bullet points)] + 1 marks (for calculations showing total marks)] + [1] mark for a bonus question. Answer all these parts together with a heading as marking scheme.

- Determine the mod-3 reduction of length 8 signal vector given as $\mathbf{x}(\mathbf{n}) = [1, 2, -2, 3, 4, -2, -1, 1]^T$. Then, sum of samples of mod-3 reduction obtained gives (?).
- Find the circular convolution of sequences $x_1(n) = [1, 2, 3, 1]$ and $x_2(n) = [1, 1, 1]$. Then, the first sample of the result gives (??).
- Find the 4-point DFT of a sequence given by $x(n) = 3\delta[n] - 2j\delta[n-1] + \delta[n-2] + 2j\delta[n-3]$. Then, $X[0] + X[2]$ gives (???)
- Bonus question: Are abrupt changes in the amplitude of a continuous, or discrete, signal associated with low or high frequencies?

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Maximum Time: 60 minutes

04/12/24

Maximum Marks: 30

1. Give a scheme to compute the 3072-point DFT of a sequence of length 3072 using 512-point FFT modules and complex multiplications and additions. Show the scheme in block-diagram form, the design should be cost effective and show complete analysis. For the design, the costs of individual modules are as given below. (15)

- one 512-point FFT module costs Rs. 500
- one multiplier costs Rs. 100
- one delay/advance unit costs Rs. 10
- one adder costs Rs. 50
- one up/down sampler unit costs Rs. 200
- If your design needs any other element, you can specify the same and add the cost as Rs. 100 per element
- the connecting wires do not add to the final cost.

2. If you are given a basic filter module (a hardware or computer subroutine), it is sometimes possible to use it repetitively to implement a new filter with sharper frequency-response characteristics. One approach, as we discussed in the lecture, and you used in the last lab exercise, is to cascade the filter with itself two or more times. Another approach, suggested by Tukey is called “twicing,” as shown in the block diagram given by Figure 1. Note that the overall frequency response is denoted as $G(e^{jw})$ (from $x[n]$ to $y[n]$).

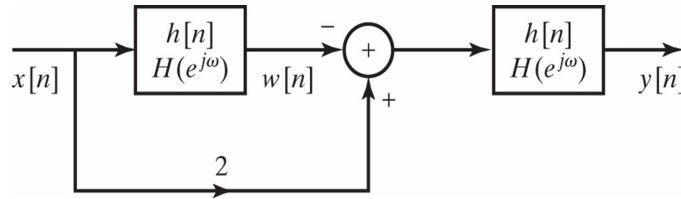


Figure 1: Twicing approach.

- (a) Assume that the basic system has a symmetric finite-duration impulse response, i.e., $h[n] = h[-n]$, $-L \leq n \leq L$. Determine whether the overall response $g[n]$ is (i) FIR and (ii) symmetric. (5)
- (b) Suppose $H(e^{jw})$ satisfies following approximation error specifications

$$(1 - \delta_1) \leq H(e^{jw}) \leq (1 + \delta_1), \quad 0 \leq w \leq w_p, \\ -\delta_2 \leq H(e^{jw}) \leq \delta_2, \quad w_s \leq w \leq \pi.$$

If the basic system has these specifications, it can be shown that the overall frequency response $G(e^{jw})$ (from $x[n]$ to $y[n]$) satisfies the specifications of the

form which are given as

$$\begin{aligned} A &\leq G(e^{jw}) \leq B, \quad 0 \leq w \leq w_p, \\ C &\leq G(e^{jw}) \leq D, \quad w_s \leq w \leq \pi. \end{aligned}$$

Determine A, B, C, and D in terms of δ_1 and δ_2 . If $\delta_1 \ll 1$ and $\delta_2 \ll 1$, what are the approximate maximum passband and stopband approximation errors for $G(e^{jw})$. (5)

- (c) You may have determined in part (b), Tukey's twicing method improves the passband approximation error, but increases the stopband error. Kaiser and Hamming generalized the twicing method so as to improve both the passband and the stopband. They called their approach "sharpening." The simplest sharpening system that improves both passband and stopband is shown in Figure 2. Assume again that the impulse response of the basic system (denoted by frequency response $H(e^{jw})$) is as given in part (a). Repeat part (b) for the system of Figure 2. (5)

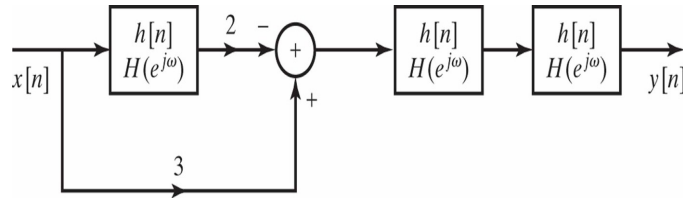


Figure 2: Sharpening approach.